Exponentials & Logarithms

ANSWER KEY

Section 1: Exponential Equations & Radicals

1 Solve the exponential equation for x: $27^{x-1} = 9^{x+2}$.

a. 5

b. 6

c. 7 ✓

d. 8

Convert to base 3: $(3^3)^{x-1} = (3^2)^{x+2} \Rightarrow 3^{3x-3} = 3^{2x+4}$. Equating exponents: $3x - 3 = 2x + 4 \Rightarrow x = 7$.

2 Simplify the radical expression: $(81x^8y^{12})^{1/4}$ for positive x and y.

a. $3x^2y^3$ ✓

b. $9x^2y^3$

c. $3x^4y^8$

d. $9x^4y^8$

Apply exponent to each factor: $81^{1/4} = 3$, $(x^8)^{1/4} = x^2$, $(y^{12})^{1/4} = y^3$. Result is $3x^2y^3$.

3 If $2^x = 5$, what is the value of 2^{x+3}?

a. 8

b. 15

c. 40 ✓

d. 125

$2^{x+3} = 2^x \times 2^3 = 5 \times 8 = 40$.

Section 2: Logarithmic Properties & Evaluation

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4 Evaluate $\log_2(64) - \log_3(27)$.

a. 1

b. 2

c. 3 ✓

d. 4

$\log_2(64) = 6$ because $2^6 = 64$. $\log_3(27) = 3$ because $3^3 = 27$. Difference $= 6 - 3 = 3$.

5 Condense into a single logarithm: $2\log x + 3\log y - \log z$.

a. $\log(\frac{x^2y^3}{z})$ ✓

b. $\log(\frac{2x \cdot 3y}{z})$

c. $\log(x^2 + y^3 - z)$

d. $\log(\frac{x^2z}{y^3})$

Apply power rule: $\log(x^2) + \log(y^3) - \log z = \log(x^2y^3) - \log z = \log(\frac{x^2y^3}{z})$.

6 If $\log_{10}(2) \approx 0.301$, what is the approximate value of $\log_{10}(20)$?

a. 1.301 ✓

b. 2.301

c. 0.602

d. 3.010

$\log_{10}(20) = \log_{10}(2 \times 10) = \log_{10}(2) + \log_{10}(10) = 0.301 + 1 = 1.301$.

Section 3: Solving Logarithmic Equations

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7 Solve for x: $\log_5(2x+7) = 2$.

a. 9 ✓

b. 11

c. 18

d. 25

Convert to exponential form: $2x + 7 = 5^2 = 25 \Rightarrow 2x = 18 \Rightarrow x = 9$.

8 Solve for x: $\log(x) + \log(x-3) = 1$ in base 10.

a. 5 ✓

b. 2

c. -2

d. 5 and -2

$\log(x(x-3)) = 1 \Rightarrow x^2 - 3x = 10 \Rightarrow x^2 - 3x - 10 = 0 \Rightarrow (x-5)(x+2) = 0$. Since $x > 3$, the only valid solution is $x = 5$.

9 What is the solution to $\ln(e^{3x-1}) = 8$?

a. 2

b. 3 ✓

c. 4

d. $\frac{7}{3}$

Using $\ln(e^u) = u$: $3x - 1 = 8 \Rightarrow 3x = 9 \Rightarrow x = 3$.

Section 4: Exponential Growth, Decay & Compound Interest

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10 A radioactive isotope decays according to $N(t) = 800(0.5)^{t/12}$, where t is measured in days. How much of the isotope remains after 36 days?

a. 50 g

b. 100 **g** ✓

c. 200 g

d. 400 g

At $t = 36$: $t/12 = 36/12 = 3$ half-lives. $N(36) = 800(0.5)^3 = 800 \times \frac{1}{8} = 100$ grams.

11 A principal of $5,000$ is invested at an annual interest rate of 4% compounded quarterly. Which expression represents the balance after 5 years?

a. $5000(1 + \frac{0.04}{4})^{20}$ ✓

b. $5000(1 + 0.04)^5$

c. $5000(1 + \frac{0.04}{12})^{60}$

d. $5000e^{0.20}$

Compound interest formula: $A = P(1 + \frac{r}{n})^{nt}$. Here $n = 4$ and $t = 5$, giving exponent $4 \times 5 = 20$.

12 A population of fish grows by 8% each year. If the initial population is $1,200$, what is the population after t years?

a. $1200(1.08)^t$ ✓

b. $1200(0.92)^t$

c. $1200(1 + 0.8)^t$

d. $1200 + 8t$

Growth rate is $r = 0.08$. The exponential model is $P(t) = P_0(1 + r)^t = 1200(1.08)^t$.

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